Answer Key — Nature of Matter: Elements, Compounds, and Mixtures

Final answers with brief reasons.

Objective

Q1	**(a) keep their own properties** — mixture components keep their properties.
Q2	**(c) oxygen** — oxygen cannot be broken down further — an element.
Q3	**(b) zinc** — brass = copper + zinc.
Q4	**(b) compound** — water is a compound of hydrogen and oxygen.
Q5	**(b) hydrogen** — hydrogen gives the pop sound.
Q6	**(b) False** — compounds split only by chemical means.
Q7	**alloy** — metal mixtures are alloys.
Q8	**milky** — it turns milky due to calcium carbonate.

Short answer

Q9	Element: a pure substance that can't be broken down (oxygen). Compound: elements chemically combined in a fixed ratio (water).
Q10	Mixture: not combined, any ratio, components keep properties. Compound: chemically combined, fixed ratio, new properties (any two).
Q11	Air's gases are merely mixed (any proportion, keep properties); water's hydrogen and oxygen are chemically combined in a fixed ratio with new properties.

Long / application

Q12	Oxygen — element (can't be broken down). Air — mixture (gases evenly mixed, any ratio). Water — compound (hydrogen + oxygen chemically combined). Brass — mixture/alloy (copper + zinc mixed).
Q13	Electricity through slightly acidified water gives hydrogen and oxygen (about 2:1 by volume); a pop test identifies hydrogen and a brighter flame identifies oxygen. Since water yields two different elements, it is a compound of hydrogen and oxygen, not an element.

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